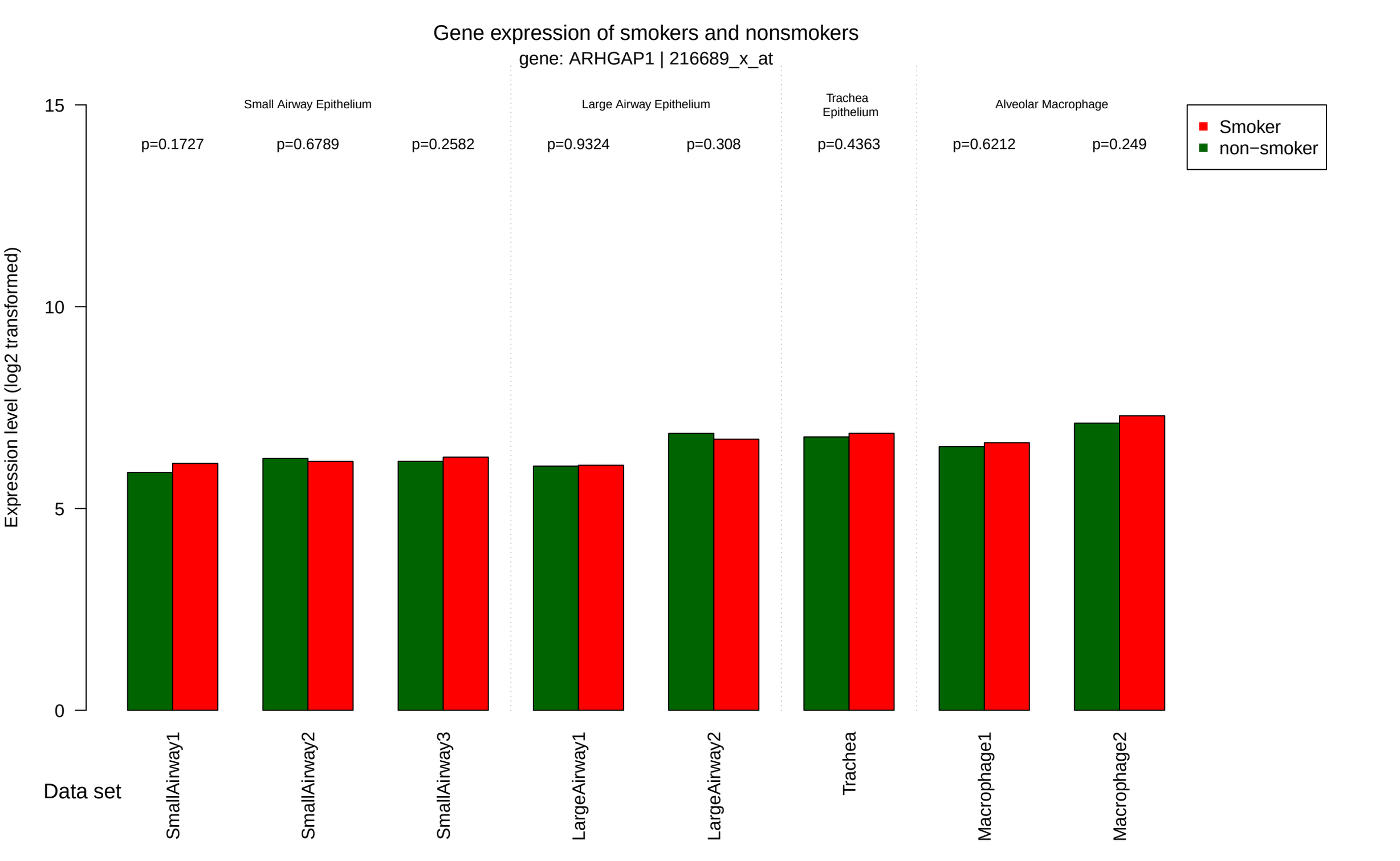
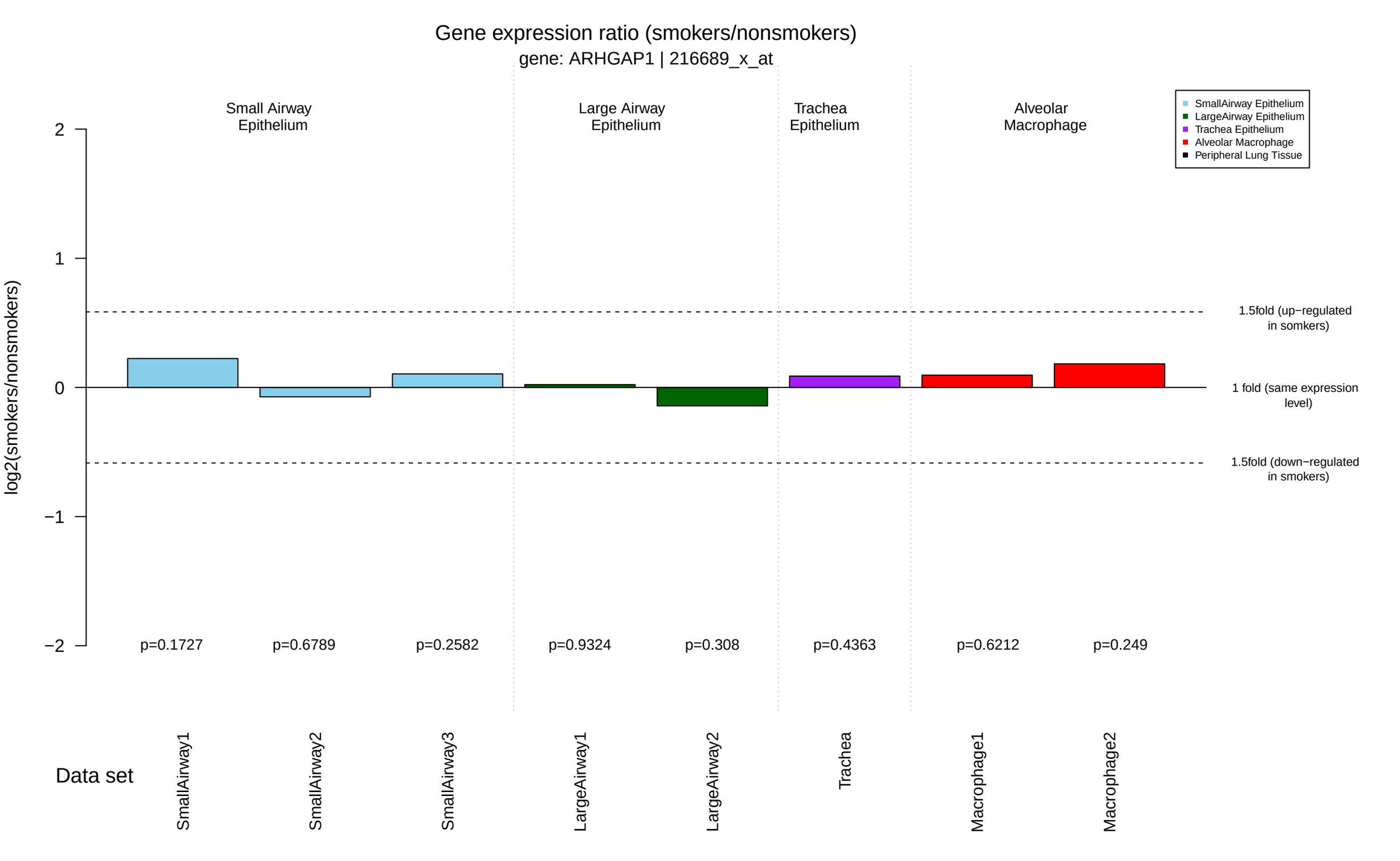
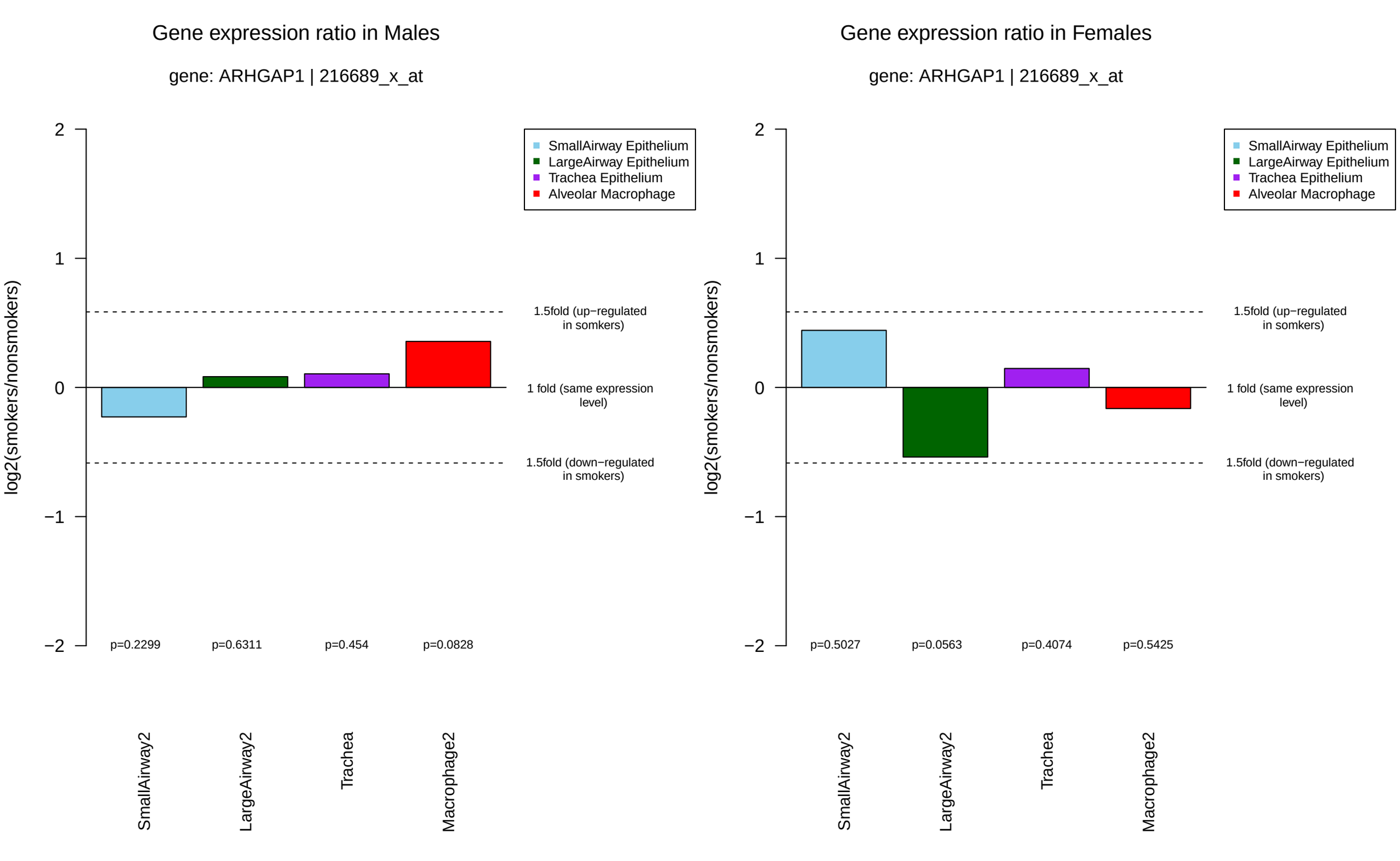
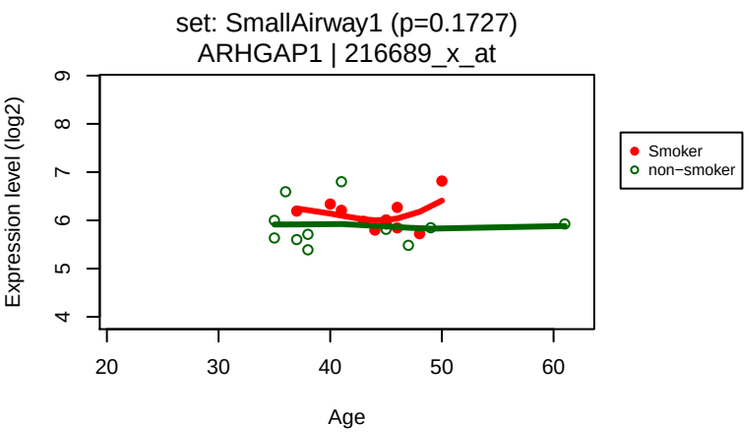
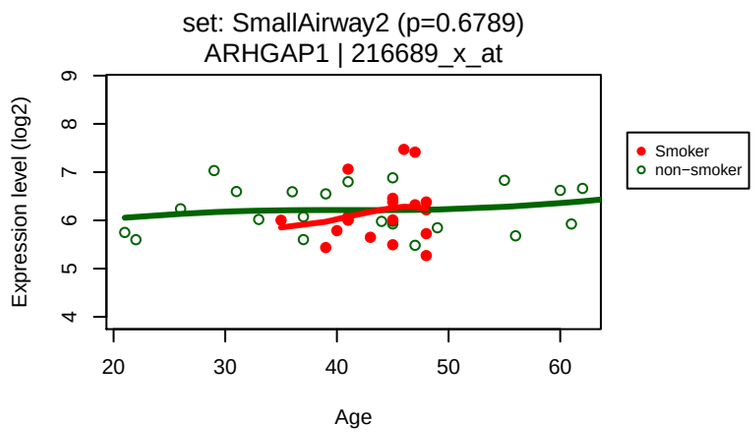
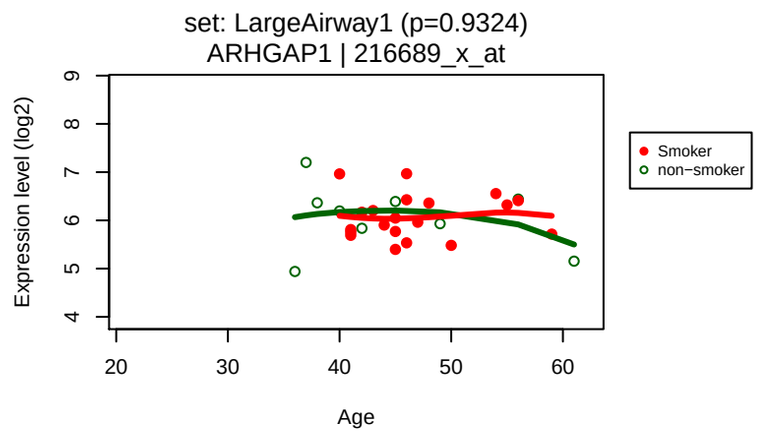

Gene expression of smokers and nonsmokers
gene: ARHGAP1 | 216689_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1727
p=0.6789
p=0.2582
p=0.9324
p=0.308
p=0.4363
p=0.6212
p=0.249
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

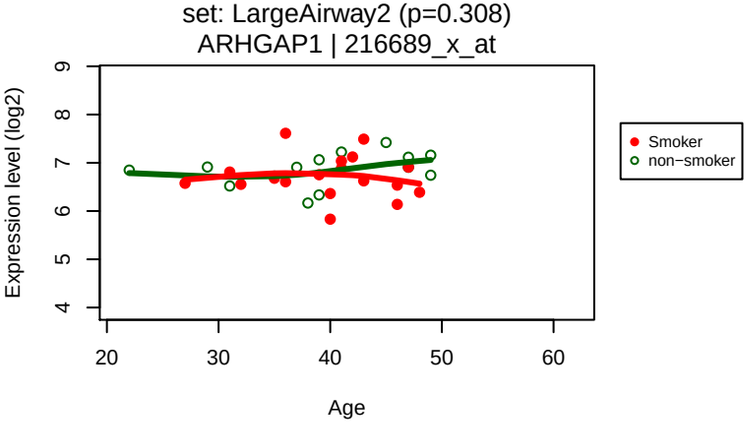

Gene expression ratio (smokers/nonsmokers)
gene: ARHGAP1 | 216689_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1727
p=0.6789
p=0.2582
p=0.9324
p=0.308
p=0.4363
p=0.6212
p=0.249
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

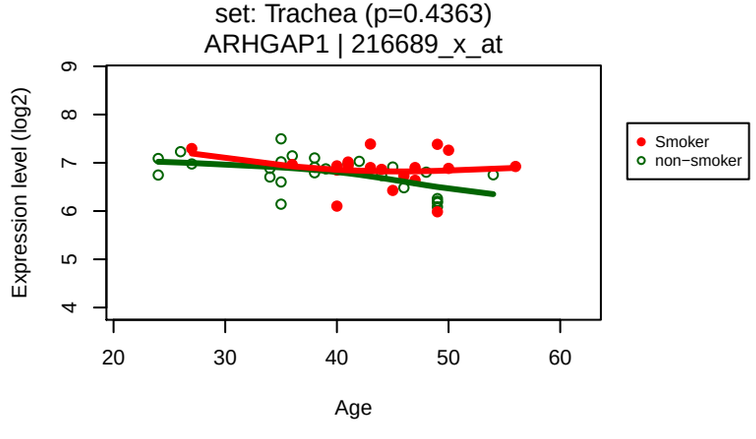

Gene expression ratio in Males
gene: ARHGAP1 | 216689_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2299
p=0.6311
p=0.454
p=0.0828
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGAP1 | 216689_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5027
p=0.0563
p=0.4074
p=0.5425
SmallAirway2
LargeAirway2
Trachea
Macrophage2

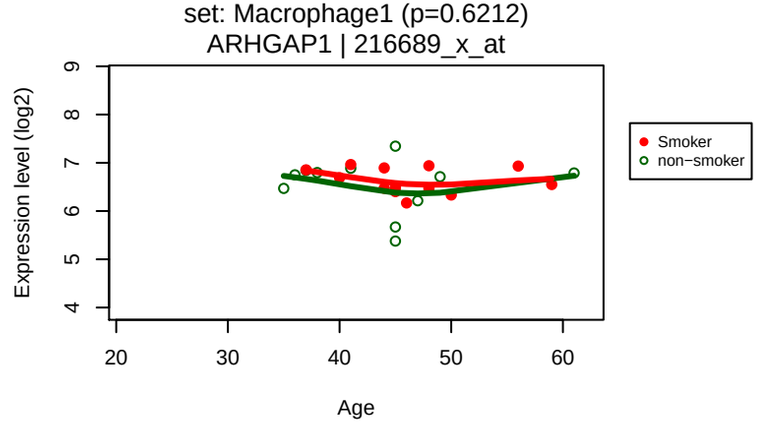

set: SmallAirway1 (p=0.1727)
ARHGAP1 | 216689_x_at
Expression level (log2)
Age
Smoker
non−smoker

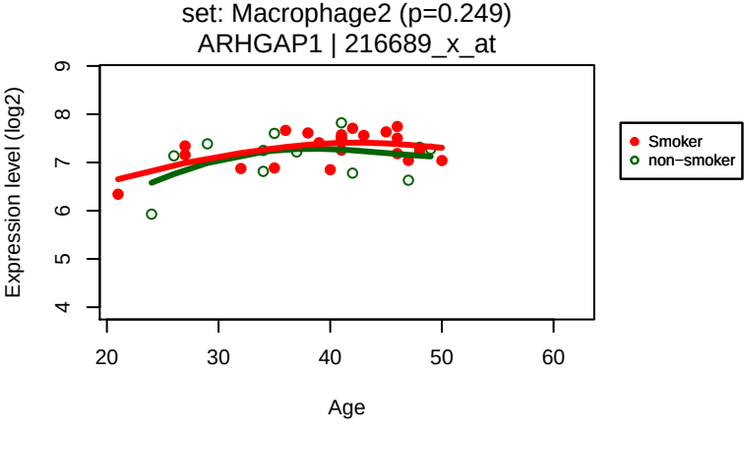

set: SmallAirway2 (p=0.6789)
ARHGAP1 | 216689_x_at
Expression level (log2)
Age
Smoker
non−smoker

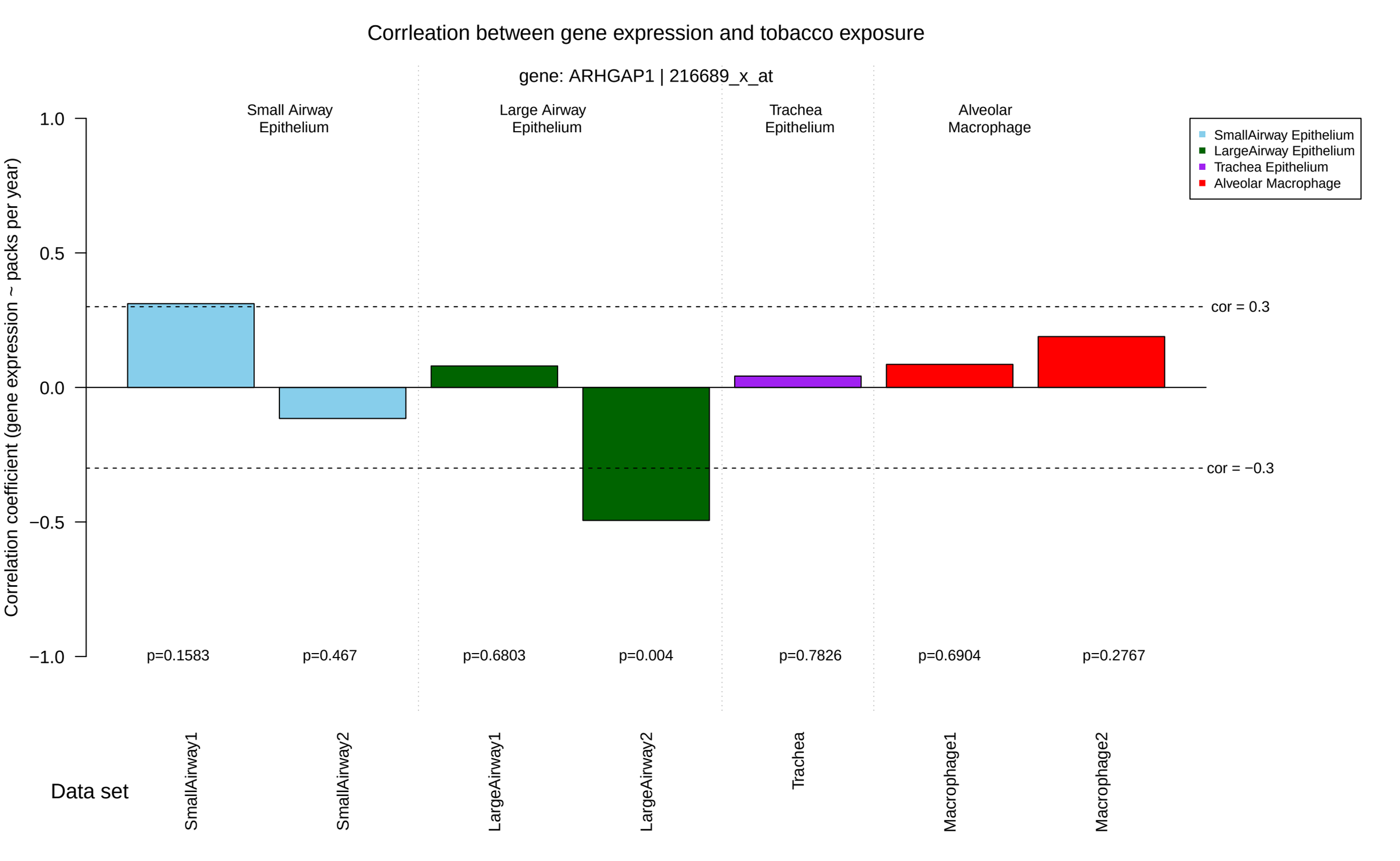

set: LargeAirway1 (p=0.9324)
ARHGAP1 | 216689_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.308)
ARHGAP1 | 216689_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4363)
ARHGAP1 | 216689_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6212)
ARHGAP1 | 216689_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.249)
ARHGAP1 | 216689_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGAP1 | 216689_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1583
p=0.467
p=0.6803
p=0.004
p=0.7826
p=0.6904
p=0.2767
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2